

# Double Pendulum and The Chaos Theory

Winston (Wenqin) Huang

Olive Tree International Academy

May-June, 2026

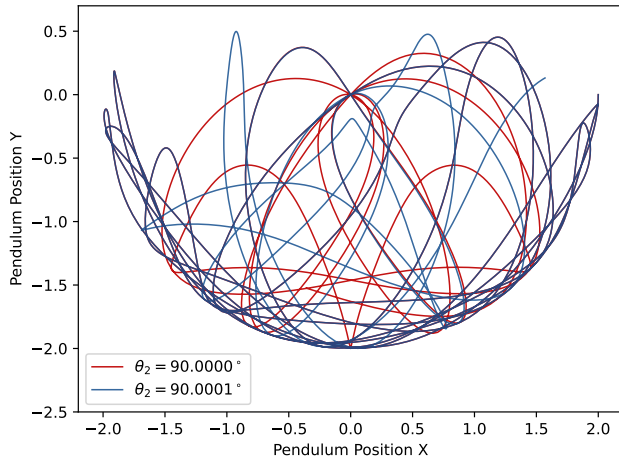
# Related CALCAB Knowledge

**Differential Equation:** It is the core of simulation algorithm we applied in this project, the entire simulation was run based on a differential equation derived by *Lagrangian Mechanics*, so the differential equation we learned in unit 7 is the core of our project.

**Limit and Approaching:** In our project, the observation of the simulation accuracy as time step changes is also a core part. By speculating the result approaches a value as step decreases (becomes more accurate), we could choose a suitable time step for simulation that is both efficient and accurate.

# Introduction To The Chaos Theory

Two Slightly Different Pendula Simulation Outcome



Double Pendulum, Where Newtonian Physics Dies.